Question:
A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient's X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient's condition?
Answer Choices:
A. It activates TLR5 on the surface of macrophages.
B. It primarily induces the Th1-cell response.
C. Peptidoglycan is its major antigen that induces an immune response.
D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

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The patient presents with symptoms of community-acquired pneumonia (CAP), including sore throat, non-productive cough, fever, oropharyngeal erythema, and patchy reticular opacities on chest X-ray. The clinical picture, along with the response to cephalexin (a broad-spectrum antibiotic effective against Gram-positive bacteria), suggests a bacterial etiology. In community-acquired pneumonia, common pathogens include Streptococcus pneumoniae, Haemophilus influenzae, and Moraxella catarrhalis. Among these, Streptococcus pneumoniae is a frequent cause and is a Gram-positive coccus with a cell wall containing peptidoglycan.

The key question is: which immune response is elicited by the pathogen?

Let âĢĻs evaluate each option:

A. It activates TLR5 on the surface of macrophages.
TLR5 recognizes flagellin from bacterial flagella. This is not relevant to the most common pathogens in CAP like S. pneumoniae, which lack flagella. Incorrect.

B. It primarily induces the Th1-cell response.
This is plausible. Bacterial pneumonias typically involve innate and adaptive immune responses, with Th1 responses being important in fighting intracellular bacterial infections (e.g., via macrophage activation and cytokine production like IFN-Î³). However, some pneumonias (especially viral) are more Th2-mediated; in contrast, bacterial pneumonia like that from S. pneumoniae generally triggers a strong Th1 response, especially with neutrophil and macrophage activation. This option is a strong candidate.

C. Peptidoglycan is its major antigen that induces an immune response.
Peptidoglycan is a major component of bacterial cell walls and is recognized by pattern recognition receptors like TLR2 and NOD1. While peptidoglycan is present in pathogens causing CAP (e.g., S. pneumoniae), the option says "peptidoglycan is its major antigen" âĢĶ this is a bit misleading. Antigens are specific proteins or molecules that elicit immune responses. While peptidoglycan induces immune responses, the question is asking about the "immune response elicited," not the antigen. Thus, this is less precise than options about cytokine responses or T-cell types.

D. This pathogen evades the immune response by encapsulation.
Encapsulation is a feature of *Streptococcus pneumoniae* and helps evade phagocytosis by masking surface antigens. However, this describes a mechanism of immune evasion, not the nature of the immune response elicited. The question specifically asks about the immune response elicited, not immune evasion.

Therefore, the immune response associated with the pathogen causing CAP (especially S. pneumoniae) is primarily a Th1 response, which helps in macrophage activation and clearance of intracellular bacteria.

<answer> B. It primarily induces the Th1-cell response. </answer>
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